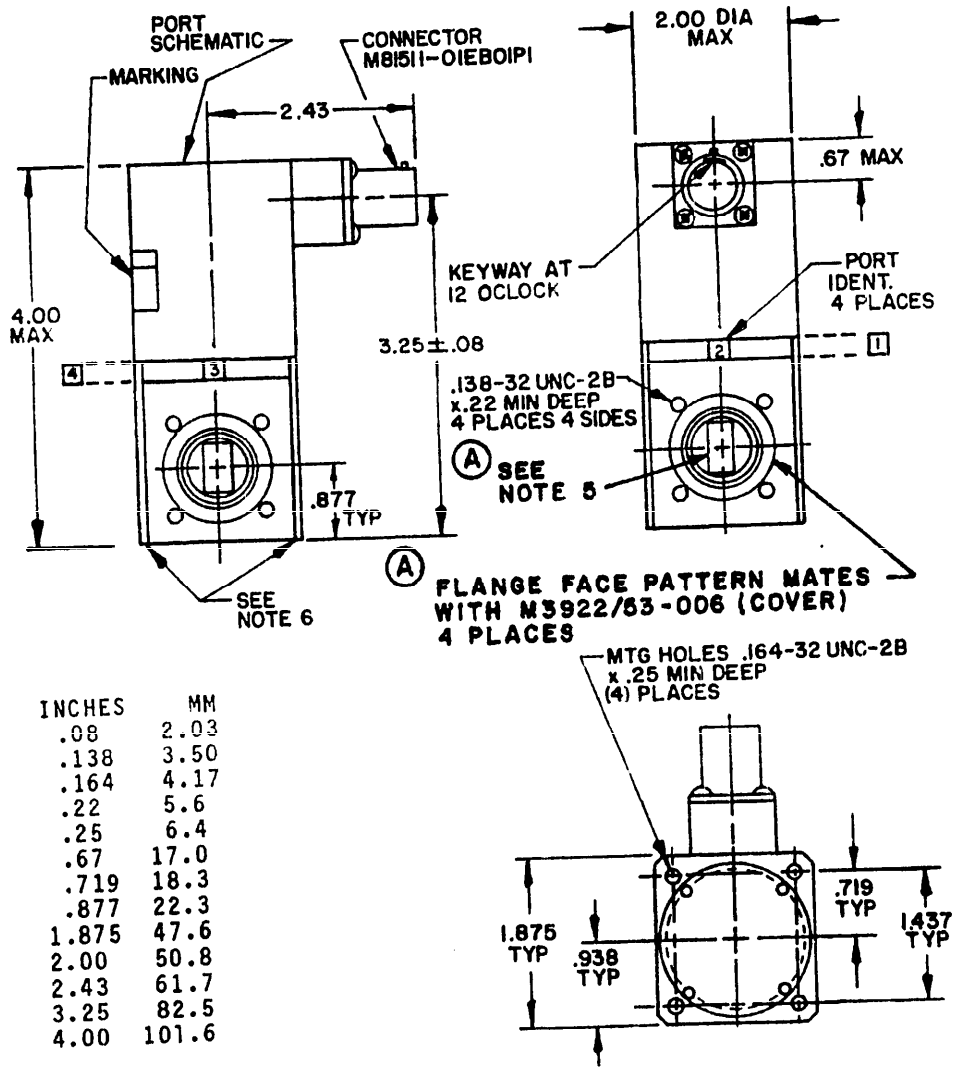
MILITARY SPECIFICATION SHEET

SWITCHES, WAVEGUIDE, TRANSFER, LATCHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-55041.



INCHES	MM
.08	2.03
.138	3.50
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.719	18.3
.877	22.3
1.875	47.6
2.00	50.8
2.43	61.7
3.25	82.5
4.00	101.6

FIGURE 1. Switch configuration and schematic, part no. M55041/4-001.

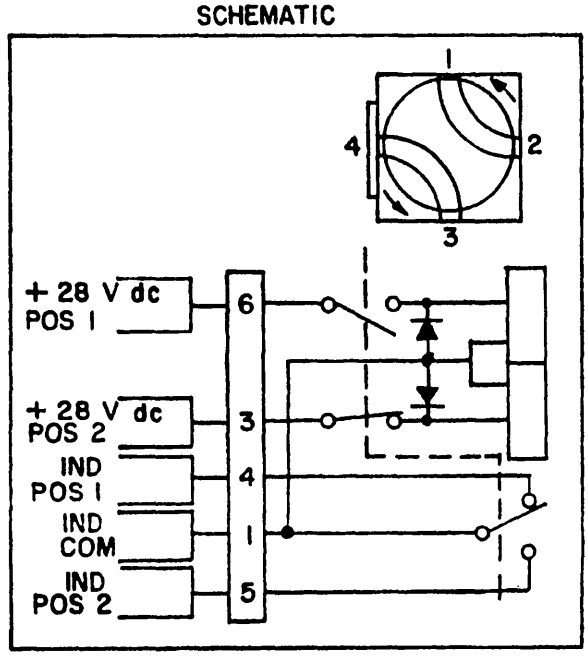
Ⓐ denotes changes

FSC 5985

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INCHES	MM
.08	2.03
.138	3.51
.164	4.17
.22	5.59
.25	6.35
.67	17.02
.719	18.26
.877	22.28
.937	23.80
1.437	36.50
1.875	47.63
2.00	50.80
2.43	61.72
3.25	82.55
4.00	101.60

SCHEMATIC

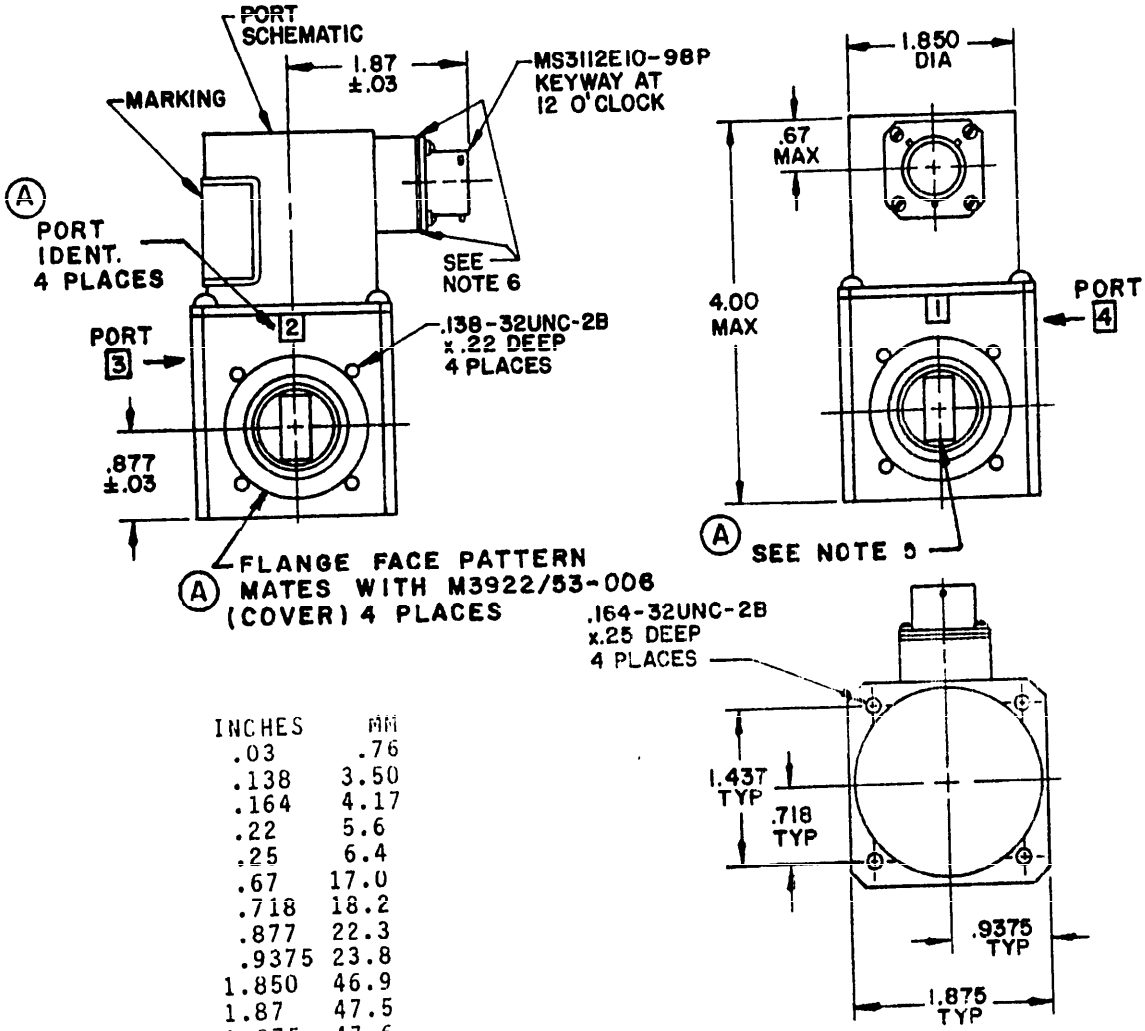


SHOWN IN POSITION 1.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
6. Painted dark gray per TT-G-527, color No. 36231 per FED-STD-595.
7. Part No. M55041/4-001 inactive for new design, use for replacement purposes only.
A 8. DC polarity must be observed.

FIGURE 1. Switch configuration and schematic, part no. M55041/4-001 - Continued.

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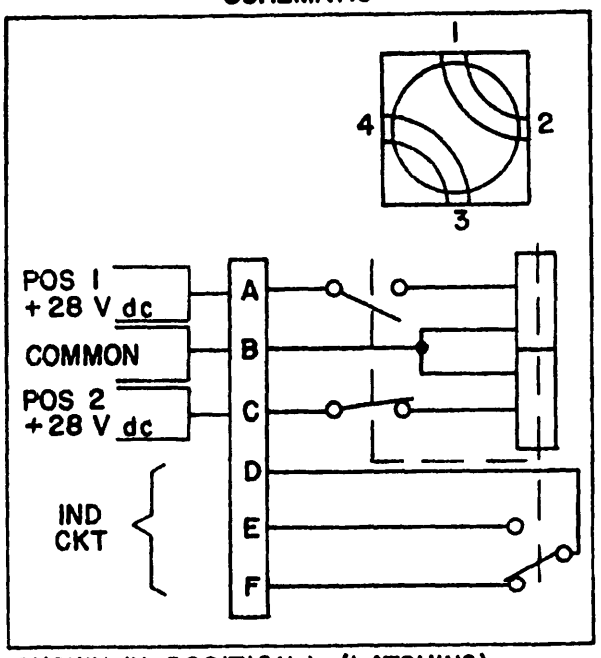
INCHES	MM
.03	.76
.138	3.50
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.718	18.2
.877	22.3
.9375	23.8
1.850	46.9
1.87	47.5
1.875	47.6
4.00	101.6

FIGURE 2. Switch configuration and schematic, part no. M55041/4-002.

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INCHES	MM
.03	.76
.138	3.51
.164	4.17
.25	6.35
.67	17.02
.718	18.24
.877	22.28
.9375	23.81
1.437	36.50
1.850	46.99
1.87	47.50
1.875	47.63
4.00	101.60

SCHEMATIC



SHOWN IN POSITION I (LATCHING)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
6. Finish: Dull black enamel.
7. DC polarity must be observed.

Ⓐ FIGURE 2. Switch configuration and schematic, part no. M55041/4-002 - Continued.

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TABLE I. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Position indicator circuit	Interlock circuit	Nominal operating voltage	Pull-in voltage (less than)	Operating current (max) 3/
			GHz	3/ amperes		V dc	V dc	amperes
001	1	1/ 2/	12.4 - 18.0	0.1	N/A	28	20	1.0
002	2	1/ 2/	12.4 - 18.0	0.1	N/A	28	20	1.0

TABLE I. Electrical performance characteristics - Continued.

Dash number	Transient interference provisions	Isolation (min)	VSWR (max)	Insertion loss (max)	Switching time	Waveguide pressurization	Life cycles (X1000)	Weight (max)
		dB		dB	ms	lb_f/in^2		oz
001	yes	40	1.1:1	0.15	150	15	200	24
002	yes	60	1.1:1	0.2	100	20	200	24

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.

2/ Rotor and housing of aluminum alloy.

3/ At 28 V dc.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

RF connectors: See figure 1.

Operation: Break before make.

Operating temperature range: -55°C to +95°C.

Electrical performance characteristics: See table I.

RF power handling capability: M55041/4-001 and M55041/4-002 = 100 W average.

Part number: M55041/4- (dash number from table I).

(A) Substitutability data: The switches covered by this specification sheet are substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

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TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985-
001	94987	120225-1	---
001	82152	33C00500	---
002	82152	33D00200	00-009-4530
002	81755	C4715-2	00-009-4530

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0975-2)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-55041/4A	Switches, Waveguide, Transfer, Latching

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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